Subject

36100 - Data Science for Innovation

8 Credit points

You are viewing the 2027 version | 2027 | Select availability

Subject description

Availabilities

Learning outcomes

Learning and teaching activities

Assessment information

Requisites

Additional subject details

Subject description

This subject introduces students to the cutting-edge initiatives that organisations and professionals are embarking on to extract and communicate value from data. The subject contextualises professional data science practices for innovation by drawing together skills and knowledge regarding data, communication, and ethics.

Students examine contemporary cases illustrating how novel data sources can act as catalysts to drive innovation and transform industries and professions globally in health, finance, insurance, management, marketing, journalism, librarianship, education, science, transportation, aviation and the environment. First-hand exploratory interaction with data enables students to engage in inquiry-based research to seek informative correlations, patterns and trends leading to discoveries, and identify data discovery studies of interest to them. Through hands-on data discovery experiments, students explore data as complex and heterogeneous, with new techniques and technologies raising implications for approaches to the ethical and legal collection of personal data.

Read Less

Faculty
TD School

Subject type
Subject

Study level
Postgraduate

Result type
Grade and marks

Total Workload hours
200

Apply now

Important dates

Fee calculator

Availabilities

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Learning outcomes

Collapse all

Upon completion of this subject, graduates will be able to:

1. Apply data summarization and preprocessing techniques to clean, transform, and prepare data for analysis, ensuring data quality and integrity.
2. Formulate and test hypotheses, interpret p-values, and make informed conclusions based on statistical evidence.
3. Utilize clustering and association rule analysis to extract insights from transdisciplinary data through innovative methods.
4. Conduct classification and regression tasks with innovative approaches, creating models capable of categorizing both structured and unstructured data into predefined classes or predicting continuous numerical outcomes.
5. Effectively communicate transdisciplinary data discovery projects tailored to the specific discipline, audience, and purpose, and critically evaluate the privacy, ethical, and legal implications of collecting and publishing personal data.

Learning and teaching activities

Blend of online and face to face activities: This subject is offered through a series of block sessions and blends online with face-to-face learning. Students participate in interactive learning experiences in timetabled on-campus sessions, where they make use of the subject materials that they have already engaged with online. In between campus sessions, students will engage in individual and collaborative online activities designed to consider a range of challenges associated with big data and innovation.

Transdisciplinary approaches: Starting from an elemental perspective on data and data science, students will approach learning from their specific professional and potential future contexts. As the subject progresses conceptual and philosophical approaches to the association between data and innovation, provocative questions will stimulate analytical engagement across a range of perspectives. Case studies and insights from industry experts will provide 'lived experiences' to accelerate and consolidate student learning.

Future-oriented strategies: Students will be exposed to contemporary learning models utilising disruptive, controversial and speculative thinking, as well as reflection. Electronic portfolios will be utilised to curate, consolidate and provide evidence of learning and development of course outcomes, graduate attributes and professional evolution. Formative feedback and suggestions for successful engagement with all assessment activities will be offered throughout the session.

Collaborative work: A strong emphasis is placed on group activities and interaction, given that graduates of this course will need to approach professional projects and challenges from a collaborative and consensus position. Individual insight is shared and reworked in groups analysing data sets for trends and assumptions. Group activities will enable students to leverage peer-learning and demonstrate effective team participation and contribution, as well as an appreciation of diverse perspectives on data science and innovation.

Weekly study and preparation activities, as well as detailed assessment information are provided in Canvas.

An aim of this subject is to help you develop academic and professional language and communication skills in order to succeed at university and in the workplace. To determine your current academic language proficiency, you are required to complete an online language screening task, OPELA. (More information is available at About OPELA for Students). If you do not complete the OPELA task, you will receive a sanction preventing access to view your subject results. If you receive a Basic result for OPELA, we will invite you to attend a Language Development Tutorial (each week from week 4 to week 11 or 12). These tutorials are designed to support you to develop your language and communication skills.

Read Less

Assessment information

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Requisites

Requisite detail

Before you enrol in this subject, you must ensure you have met its requisites. The requisites for this subject can be viewed on its access conditions page.

To learn more about requisites, please visit the How to use the Handbook guide.

Additional subject details

Collapse all

Attendance requirement

Scheduled synchronous sessions (online or on campus) and participation in online activities

UTS acknowledges the Gadigal people of the Eora Nation, the Boorooberongal people of the Dharug Nation, the Bidiagal people and the Gamaygal people, upon whose ancestral lands our university stands. We would also like to pay respect to the Elders both past and present, acknowledging them as the traditional custodians of knowledge for these lands.

The page is authorised by the Provost, UTS

© Copyright UTS - TEQSA

Provider ID: PRV12060 - TEQSA
Category: Australian University

CRICOS Provider No: 00099F

ABN: 77 257 686 961